PRE-SEED · $5M TARGET · PLANNED SAFE

Make change the unit of compute

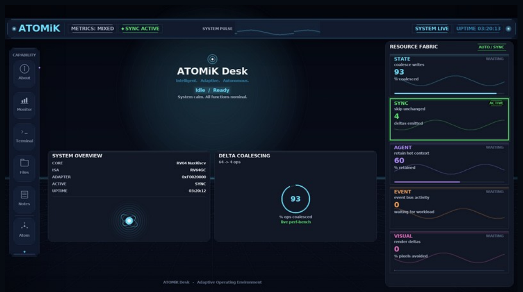
For edge and embedded teams whose battery, heat, bandwidth, latency, or footprint is constrained by repeated state work.

1
workload
Start with one constrained state path.

1
baseline
Measure against the current system.

1
constraint
Battery, heat, latency, bandwidth, footprint, or hardware limit.

$5M
ask
18-24 month proof round. No tape-out funding.



Current Zynq UI proof image - caption-bound, not a customer benchmark

PROBLEM

Customers already know the pain

- Batteries die too quickly.
- Hardware is maxed out.
- Systems run hot.
- Devices are too big or too heavy.
- Costs keep increasing.
- More performance is needed.

Before they spend more money

- bigger hardware
- larger batteries or cooling
- more bandwidth or a redesign

ATOMiK checks whether a constrained state path can get more out of the system they already have.

Customer-language pain framing. Outcomes remain workload-specific until measured.

END-GAME VALUE

Customers do not buy delta-state algebra

Bytes	Latency	Ops	Power
avoided	**reduced**	**coalesced**	**proxied**
less state movement against the baseline	faster local state paths where fit exists	less repeated work in change-heavy paths	follow-on measurement when instrumentation exists

Customer ROI is measured after the workload shows reduced state movement while preserving correctness.

Battery, heat, cooling, water, footprint, and power-bill outcomes remain evaluation targets until measured end to end.

MARKET OPPORTUNITY

Large pressure pool, focused wedge, strategic optionality

$1T+

TAM context

2026 semiconductor revenue context. Backdrop only; ATOMiK does not address all semiconductor spend.

$112B-$169B

SAM context

Embedded systems market range from 2024 estimate to 2030 forecast; edge AI adds adjacent pressure.

$10M-$40M

entry wedge

early annual revenue path from evaluations, design partners, and licensing; not a ceiling.

Venture logic: proof-backed IP can support licensing and strategic optionality if workload evidence and partner diligence hold.

Sources: SIA/WSTS, Gartner, Grand View Research, MarketsandMarkets. Semiconductor IP: $8.14B 2025 to $11.2B 2029. Planning targets are not forecasts.

FOCUS MARKETS

Growth pressure meets constrained state work

Edge AI

$24.9B 2025 -> $118.7B 2033
21.7% CAGR

Target examples: NVIDIA Jetson ecosystem, Qualcomm, Ambarella, Hailo, Advantech

Robotics / industrial

$33.8B 2024 -> $131.5B 2030
26.5% CAGR

Target examples: Rockwell Automation, Siemens, ABB, FANUC, Universal Robots

Drones / remote autonomy

$30.0B 2024 -> $54.6B 2030
10.6% CAGR

Target examples: Skydio, AeroVironment, Anduril, Shield AI, Teledyne FLIR

Smallsat / remote sensing

$4.0B 2024 -> $14.0B 2030
22.8% CAGR

Target examples: Planet, Spire, BlackSky, Maxar, Rocket Lab

Sources: Grand View Research segment reports. Target examples are account hypotheses only; no relationship or endorsement implied.

COMPETITIVE LANDSCAPE

Alternatives are real; proof-bound IP is the wedge

More silicon / accelerators

NVIDIA Jetson, Qualcomm, Ambarella, Hailo, Google Coral; FPGA upgrades from AMD/Xilinx, Intel/Altera, Lattice

Processor and IP incumbents

Arm, Synopsys ARC, Cadence Tensilica, CEVA, Andes/RISC-V ecosystem, Imagination

Software/status quo

compression, caching, dedup, delta sync, CRDT/event sourcing, hand optimization, bigger batteries, more cooling

Defensibility wedge

Lean4-checked formal algebra + FPGA hardware validation + IP/proof registry. Exact formal claims stay tied to audited properties.

Competitor names are buyer alternatives/status quo references, not partnership, customer, or displacement claims.

PRIMITIVE

Make change the unit of compute

state = reference_state XOR accumulated_delta

LOAD
set reference

ACCUM
add deltas

READ
reconstruct

SWAP
commit boundary

Plain English: track meaningful change and reconstruct only when the system needs full state.

FIT

ATOMiK has a wedge, not a universal claim

- updates dominate reads
- state changes are sparse
- bandwidth or latency is expensive
- power budget or hardware margin is tight
- sync, replay, rollback, or context retention creates overhead

Not the claim

- CPU/GPU/NPU replacement
- universal speedup
- thermal or battery result before measurement

USE CASES

Same waste, different customer budgets

Robotics / industrial

heat, latency, reliability, control-state churn

AI at the edge

context movement, memory pressure, local response

Remote systems

drones, satellites, field runtime, packet budget

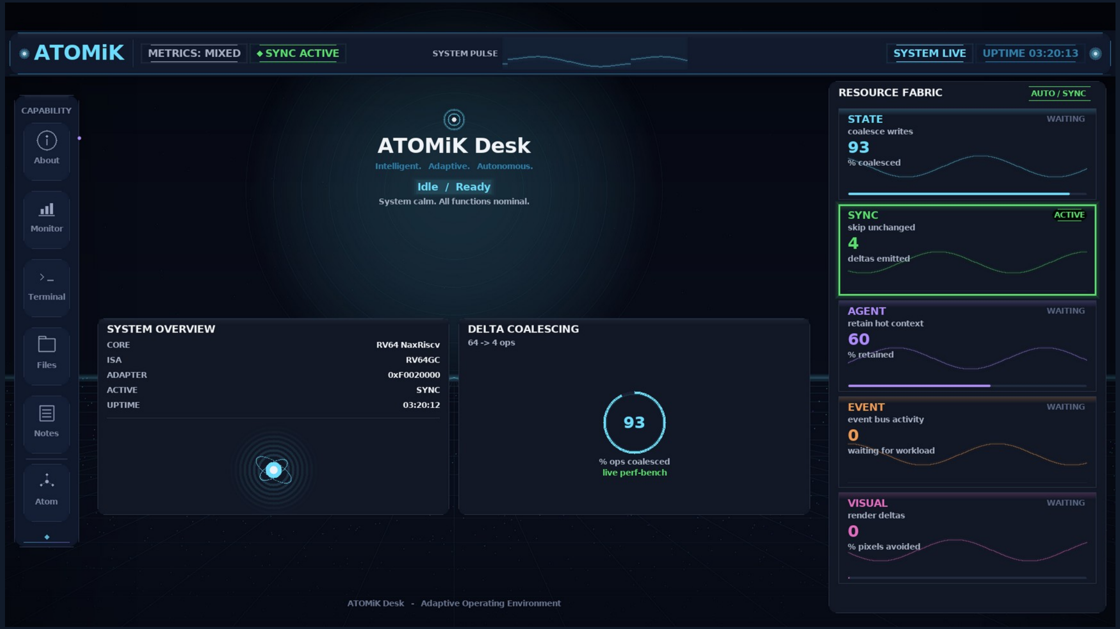
Defense-adjacent

constrained hardware, power, rugged reliability

ICP is defined by constraint and measurable state path, not by industry label alone.

LIVE PROOF

Current Zynq UI artifact with caption



What it shows

current Zynq UI captured from /dev/fb0, driven by real measured on-board data

What to quote

caption-bound hardware-validated UI artifact

What not to quote

on-screen telemetry as customer benchmark; not an interactive demo

HARDWARE_VALIDATED UI ARTIFACT - not a customer workload benchmark or production-readiness claim.

PROOF STACK

Real, specific, evidence-bounded

Zynq Desk v0.40-A	HARDWARE_VALIDATED	current UI proof image, captured from /dev/fb0
Live Workloads demo	LIVE_MEASURED	first customer-facing workload on AX7020: 800 Mevents/s @ 8 banks, byte-identical
Parallel-bank throughput	LIVE_MEASURED	AX7020 8-bank accumulator scales 1/2/4/8x, byte-identical to software
Linux userspace to FPGA	HARDWARE_VALIDATED	documented OS-to-bus path
AX7020 board matrix	LIVE_MEASURED	wins and losses with caveats
Formal algebra	FORMAL_PROOF*	Lean4-checked algebra; exact audited properties only
Standalone boot	BUILD_ARTIFACT	not promoted until run proof

*FORMAL_PROOF applies only to specifically audited formal properties; it does not imply workload or customer outcomes.

COMMERCIAL PATH

A customer can understand the first step

1. Show us the problem path

the part causing battery, heat, hardware, latency, size, weight, or cost pressure

2. We evaluate fit

measure whether ATOMiK can reduce state work against the current baseline

3. We show the evidence

where the opportunity is, what was measured, and the potential impact

4. Or call no-fit

if ATOMiK does not help, say so and stop before wasting customer time

Design-partner work and licensing diligence come only after measured proof supports the path.

FINANCIAL MODEL

$5M funds proof, partners, and licensing diligence

Founder-prepared planning scenario, not a forecast.

$0-$150K	$250K-$750K	$1M-$3M	$3M-$8M
0-6 mo	**6-12 mo**	**12-24 mo**	**24-36 mo**
qualified proof reviews / technical evals	paid evaluation or design-partner SOWs	potential contracted value if proof converts	annualized potential if proof and economics hold

Use of funds: $1.5M engineering | $1.0M customer proof | $750K IP/legal | $750K ASIC feasibility | $1.0M finance/GTM/ops + reserve

Do not present ranges as booked revenue, committed pipeline, forecast, or guaranteed ROI. Material revenue requires signed SOWs or licensing terms.

RISKS AND GATES

Trust comes from saying what is still unproven

Customer validation

pending; design partners are the wedge

Downstream outcomes

battery/heat/water/ROI need end-to-end measurement

ASIC economics

external feasibility review before tape-out

Incumbents

differentiate by proof, IP, and workload focus

The risks are exactly why this is a proof round, not a tape-out round.

TEAM

Founder-led proof with commercial translation

Matthew H. Rockwell

Founder & CEO

Architecture, live proof, product direction, financial model, ask, and technical Q&A.

Allison Rossi

CMO

Messaging, positioning, buyer-language, commercialization, and go-to-market support.

Advisor priorities

ASIC/EDA feasibility reviewer | formal-methods reviewer | semiconductor IP/licensing counsel

No advisors are represented as signed until documented. These are priority asks for the round and AAN network.

THE ASK

$5M target to reach measured workload proof

- $5M target pre-seed; final SAFE terms CFO/counsel pending
- 18-24 months to measured workload proof, IP packet, ASIC feasibility
- Need: capital, qualified workloads, design-partner introductions

Priority help

not just capital

design-partner introductions plus investor feedback

Money in

$5M target

Output

proof + IP + feasibility

Upside

licensing path if proof holds

Final SAFE mechanics CFO/counsel pending. No signed traction, revenue, or customer-savings claim without supporting documents.